
AN0003- EC NB-IoT COAP Application Note_V1.2

EigenCOMM Wireless Microcontroller

Document Description

This document describes in detail the COAP AT commands of eigencomm NB-IoT chip. It aims to help users quickly create COAP client, send and receive data, and delete COAP client through AT commands.

Contents

1.	Introduction	3
1.1	COAP protocol introduction	3
1.2	Document Purpose	4
1.3	Content Validity	4
2.	COAP Server	4
3.	Eigencomm NB-IoT Chip Application.....	4
3.1	Interact with COAP server	4
3.2	COAP AT Commands Details.....	6
3.2.1	AT+COAPCREATE	6
3.2.2	AT+COAPDEL.....	7
3.2.3	AT+COAPADDRES.....	7
3.2.4	AT+COAPHEAD	8
3.2.5	AT+COAPOPTION.....	8
3.2.6	AT+COAPSEND.....	10
3.2.7	AT+COAPDATASTATUS.....	11
3.2.8	AT+COAPCFG.....	11
3.2.9	AT+COAPALISIGN	12
3.2.10	+COAPURC	13
4.	Appendix	14
4.1	Power on inspection process	14
4.2	Interaction process with COAP server.....	14
5.	Version.....	15
6.	About US.....	16

1. Introduction

1.1 COAP protocol introduction

COAP (constrained Application Protocol) is a web like protocol in the world of the Internet of things. Its detailed specification is defined in RFC 7252. The name of COAP is translated as "limited application protocol". As the name implies, it is used on the Internet of things with limited resources.

COAP protocol features:

- 1) The network transport layer of COAP protocol is changed from TCP to UDP.
- 2) It is based on rest. The server's resource address has the same URL format as the Internet, and the client also has post, Get, put and delete methods are used to access the server to simplify http.
- 3) COAP is in binary format, HTTP is in text format, and COAP is more compact than http.
- 4) Lightweight, the minimum length of COAP is only 4b, and HTTP header has dozens of B.
- 5) Support reliable transmission, data retransmission, block transmission. Ensure reliable data arrival.
- 6) It supports IP multicast, that is, it can send requests to multiple devices at the same time.
- 7) Non long connection communication is suitable for low-power Internet of things scenarios.

The COAP protocol has four message types:

1) Con -- a request that needs to be acknowledged. If a con request is sent, the other party must respond. This is kind of like TCP, the other party must confirm to receive the message for reliable message transmission.

2) Non - a request that does not need to be acknowledged. If a non request is sent, the other party does not have to respond. This applies to

Messages will be sent repeatedly and frequently, and packet loss will not affect normal operation. This is very similar to UDP. Used for unreliable message transmission.

- 3) ACK -- reply message, corresponding to the reply of con message.
- 4) Rst -- reset message. When the received message is unknown or wrong during reliable transmission, the ACK message cannot be returned, and the rst message must be returned.

Requests and responses of COAP:

- 1) Get method -- used to get a resource
- 2) Post method -- used to create a resource
- 3) Put method -- used to update a resource
- 4) Delete method -- used to delete a resource

1.2 Document Purpose

This paper describes how to use the COAP AT commands of eigencomm NB-IoT chip to realize the customer's device as the COAP client, access the COAP server, and complete various basic interactive operations.

1.3 Content Validity

This paper first introduces the whole process of the COAP AT commands, and then enumerates the process of accessing the COAP server.

UART communication is used between MCU and eigencomm NB-IoT chip. The format of communication is AT commands. This chip supports the use of AT commands to realize all functions of COAP, including creating COAP, deleting COAP, configuring COAP parameters, sending data, receiving data.

The AT commands of the chip supports multiple platforms and connects different platforms through command parameters

2. COAP Server

Coap.me is the COAP server recommended by coap.technology. COAP related tools are introduced in coap.technology/tools.html.

When debugging, use the AT commands to communicate with the COAP server, coap.me. The main work is to create a COAP client

And delete. Send and receive the COAP data. Send the request cancellation to the coap: // coap.me: 5683 / test URI and the COAP client

The COAP server will reply to the COAP client.

It should be noted that before creating a COAP client, you need to use at + cmdns to obtain the IP address of the COAP server, because if you directly use URL, it will take too long to resolve DNS.

3. Eigencomm NB-IoT Chip Application

3.1 Interact with COAP server

The ATs commands of COAP is designed according to the functional division of COAP, including 10 AT commands: COAP create, COAP delete, COAP add resource, COAP head, COAP option, COAP status, COAP config, COAP Ali sign, COAP send and COAP receive.

The purpose of COAP create is to create socket connection and COAP client.

COAP delete is used to delete COAP, including COAP connection and socket connection.

The role of COAP send is to send data to the cloud platform. According to the COAP protocol, in the case of bad network state, COAP may

It will resend data, so when using the COAP send command, it is possible for COAP to receive multiple data from the server, that is, multiple "+ coapurc:" will be displayed.

The function of COAP add resource is to add resource. This command has not been strictly tested and is not supported temporarily.

The role of COAP COAP head is to configure the ID and token of COAP head.

The role of COAP option is to configure various options of COAP.

The purpose of COAP status is to query the sending status of COAP data.

COAP config is used to configure the display options when receiving data.

The function of coalali sign is to calculate the sign value of Ali.

The purpose of the COAP receive is to receive the data sent by the COAP server.

The flow of AT commands using COAP is as follows:

Step 1: use at + cmdns to obtain the IP address of the COAP server.

Step 2: use COAP create to create a COAP client.

Step 3: use COAP option to configure the option.

Step 4: use COAP send to send the data.

Step 5: when the COAP server issues data, the COAP receive command will be automatically received and printed out in the format of "+ coapurc:".

Step 6: use COAP delete to delete the COAP connection.

The following is a schematic diagram of interaction between eigencomm NB-IoT chip and COAP server through AT commands:

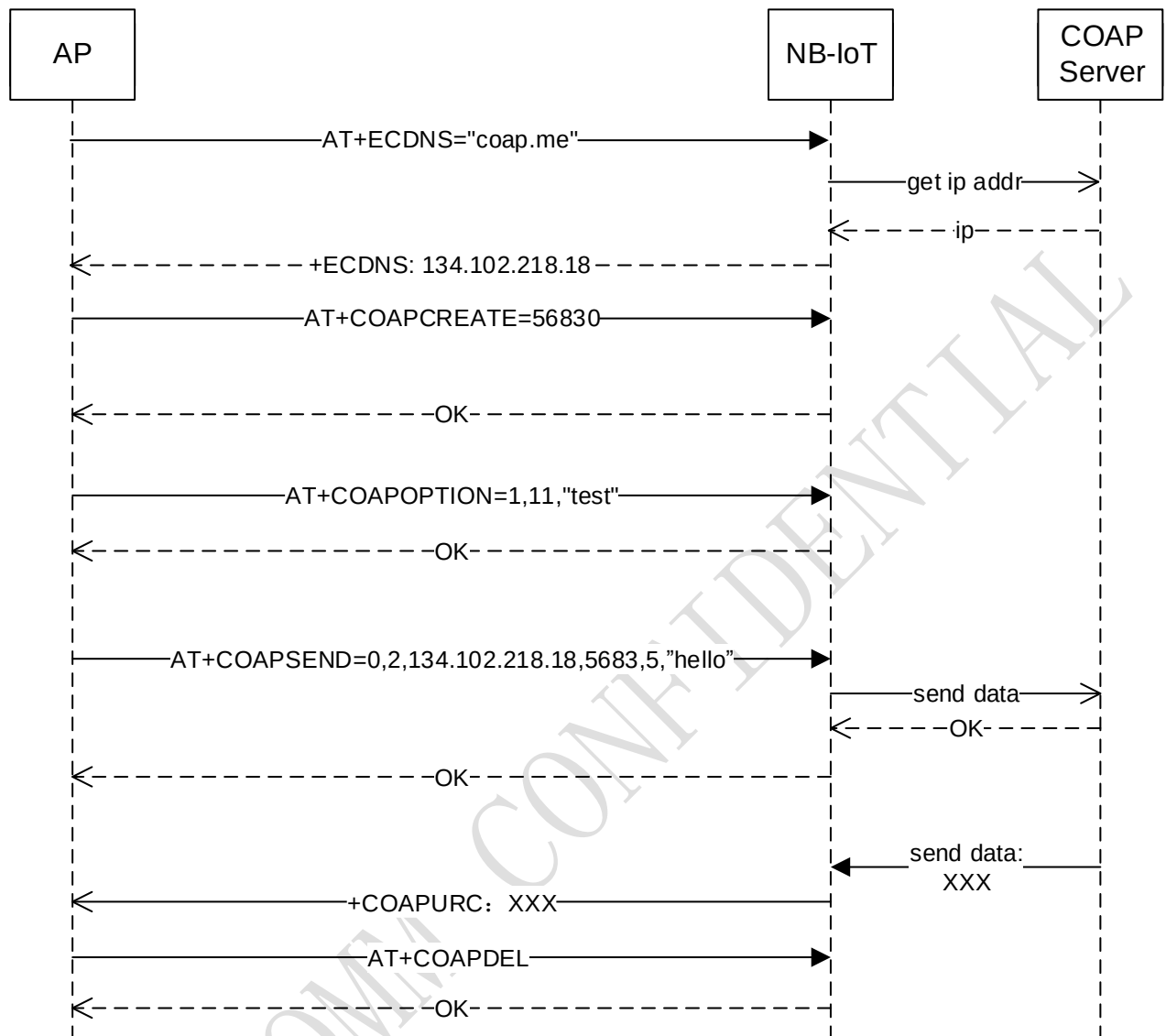


Figure1: interaction diagram

3.2 COAP AT Commands Details

3.2.1 AT+COAPCREATE

The command creates a CoAP client.

AT+COAPCREATE

Set Command AT+COAPCREATE=<local port>	Response OK If there is any error, response: +CME ERROR: <err>
Test Command	Response

AT+COAPCREATE=?	+COAPCREATE:<1-65535> OK If there is any error, response: +CME ERROR: <err>
Maximum Response Time	10s
Parameter Saving Mode	NO_SAVE

Parameter

<localport>	Integer type; client's port, values of 1-65535 are supported
-------------	--

3.2.2 AT+COAPDEL

The command deletes the CoAP client

AT+COAPDEL

Execution Command AT+COAPDEL	Response OK If there is any error, response: +CME ERROR: <err>
Maximum Response Time	5s
Parameter Saving Mode	NO_SAVE

3.2.3 AT+COAPADDRES

The command adds the CoAP resource

AT+COAPADDRES

Set Command AT+COAPADDRES=<length>,<resource>	Response OK If there is any error, response: +CME ERROR: <err>
Test Command AT+COAPADDRES=?	Response +COAPADDRES: <1-50>,"<resource>" OK If there is any error, response: +CME ERROR: <err>
Maximum Response Time	10s
Parameter Saving Mode	NO_SAVE

Parameter

<length>	Integer type The CoAP client resources, range: 1-50
<resource>	String type

The resource name

Note: this command is not supported now.

3.2.4 AT+COAPHEAD

The command adds the CoAP head

AT+COAPHEAD

Set Command AT+COAPHEAD=<mode>[, [<msgid>] [, <tkl>, <token>]]]	Response OK If there is any error, response: +CME ERROR: <err>
Test Command AT+COAPHEAD=?	Response +COAPHEAD: <mode>[, [<msgid>] [, <tkl>, <token>]]] OK If there is any error, response: +CME ERROR: <err>
Maximum Response Time	10s
Parameter Saving Mode	NO_SAVE

Parameter

<mode>	Integer type, The CoAP Head and Token parameter 1, Generate message id and token values randomly 2, Generate message id, and configure the token values 3, Only configure message id, not needed token values 4, Configure message id, and generate the token values randomly 5, Configure message and token values
<msgid>	Integer type The message id, only needed configure when the <mode> value is 3, 4, 5. Range value: 0-65535
<tkl>	Integer type The token values length, only needed configure when the <mode> value is 1, 2. Range value: 1-8.
<token>	String type The token values, hexadecimal format string, only need configure when the <mode> value is 1, 2.

3.2.5 AT+COAPOPTION

The command adds the CoAP option

AT+COAPOPTION

Set Command AT+COAPOPTION= <opt cnt>,<opt name>,<opt value>[,...]	Response OK If there is any error, response: +CME ERROR: <err>
Test Command AT+COAPOPTION=?	Response +COAPOPTION: <opt cnt,<opt name>,"<opt value>"[,...] OK If there is any error, response: +CME ERROR: <err>
Maximum Response Time	10s
Parameter Saving Mode	NO_SAVE

Parameter

<opt cnt>	Integer type The option parameter count, range value: 1-12
<opt name>	String type, The option name, refer the RFC 7252
	1 If-Match
	3 Uri-Host
	4 ETag
	5 If-None-Match
	6 Observe
	7 Uri-Port
	8 Location-Path
	11 Uri-Path
	12 Content-Format
	14 Max-Age
	15 Uri-Query
	17 Accept
	20 Location-Query
	23 Block2
	27 Block1
	28 SIZE
	35 Proxy-Uri
	39 Proxy-Scheme
	60 Size1
<opt value>	String type, The length of value string: 1-180. If the <opt_name> is 12 or 17, the <opt_value> should be the below value
	"0", text-plain
	"40", application/link-format
	"41", application/xml
	"42", application/octet-stream
	"47", application/exi

"50", application/json

3.2.6 AT+COAPSEND

The command send data to CoAP server.

AT+COAPSEND

Set Command AT+COAPSEND= <msgType>,<method>, <ipAddr>,<port>[,<length>,<data>]	Response OK If there is any error, response: +CME ERROR: <err>
Set Command AT+COAPSEND= <msgType>,<method>, <ipAddr>,<port> Note: After ">" is responded, input the data to be sent. Tap "CTRL+Z" to send, and tap "ESC" to cancel the operation	Response OK If there is any error, response: +CME ERROR: <err>
Test Command AT+COAPSEND=?	Response +COAPSEND: <msgType>,<method/rspcode> ,<ipAddr>,<port> [,<length>,<data>] OK If there is any error, response: +CME ERROR: <err>
Maximum Response Time	10s
Parameter Saving Mode	NO_SAVE

Parameter

<msgType>	Integer type
	0 CON, confirmable message(requires ACK/RST)
	1 NON, non-confirmable message(one-shot message)
	2 ACK, used to acknowledge confirmable message
	3 RST, reset, indicates error in received message
<method>	Integer type
	1 GET
	2 POST
	3 PUT
	4 DELETE
<ipAddr>	String type
	The CoAP Server ip address
<port>	Integer type
	The CoAP Server Port

<length>	Integer type
	The length of data to be sent, the max length is 512 Bytes
<data>	String type
	The length of data to be sent, hex string

3.2.7 AT+COAPDATASTATUS

The command gets the CoAP status

AT+COAPDATASTATUS

Test Command AT+COAPDATASTATUS=?	Response +COAPDATASTATUS:<status> OK If there is any error, response: +CME ERROR: <err>
Maximum Response Time	5s
Parameter Saving Mode	NO_SAVE

Parameter

<status>	Integer type
0	Have not sent
1	Sent, waiting response of IoT platform(not supported)
2	Sent failed(not supported)
3	Timeout(not supported)
4	Success
5	Got reset message(not supported)

3.2.8 AT+COAPCFG

The command configs the CoAP client

AT+COAPCFG

Set Command AT+COAPCFG="Showra" [, <Showra>]	Response OK If there is any error, response: +CME ERROR: <err>
Set Command AT+COAPCFG="Showrspopt" [, <Showrspopt>]	Response OK If there is any error, response: +CME ERROR: <err>
Test Command AT+COAPCFG=?	Response OK

Read Command AT+COAPCFG?	Response OK
Maximum Response Time	10s
Parameter Saving Mode	NO_SAVE

Parameter

<Showra>	Integer type, Set whether or not to display the address of sender
<Showrspopt>	Integer type, Set whether or not to display the coap option of sender

3.2.9 AT+COAPALISIGN

The command gets the ali cloud sign

AT+COAPALISIGN

Set Command AT+COAPALISIGN=" <devId> ", " <devName> ", " <devSecret> ", " <productKey> "	Response +COAPALISIGN: " <sign> " OK If there is any error, response: +CME ERROR: <err>
Test Command AT+COAPALISIGN=?	Response +COAPALISIGN:" <dev_id> ", " <dev_name> ", " <dev_secret> ", " <product_key> " OK If there is any error, response: +CME ERROR: <err>
Maximum Response Time	10s
Parameter Saving Mode	NO_SAVE

Parameter

<devId>	String type Device ID issued by AliCloud.
<devName>	String type Device name issued by AliCloud.
<devSecret>	String type Device secret key issued by AliCloud
<productKey>	String type Product key issued by AliClode
<sign>	String type The calculated sign value

3.2.10 +COAPURC

This is an unsolicited message to indicate CoAP client receive data from CoAP server.

+COAPURC

```
+COAPURC:"rsp", [<ip_addr>,<port>,<type>,<rspcode>,<msgid>[,<opt_cnt>,<opt_name>,<opt_value>"] [,<length>,<data>]
+COAPURC:"req", [<ip_addr>,<port>,<type>,<method>,<msgid>,<mode>[,<tkl>,<token>] [,<opt_cnt>,<opt_name>,<opt_value>"] [,<length>,<data>]
```

Parameter

<ip_addr>	String type The CoAP server ip address, it will show when set AT+COAPCFG="Showra",1
<port>	Integer type The CoAP server port, it will show when set AT+COAPCFG="Showra",1
<type>	Integer type The CoAP Protocol of message type, range: 0-3, refer the RFC 7252
<rspcode>	String type The response code of CoAP Protocol. Refer to the RFC 7252
<method>	Integer type The method of CoAP Protocol. Refer to the RFC 7252 1 GET 2 POST 3 PUT 4 DELETE
<msgid>	Integer type The CoAP message id
<mode>	Integer type Indicates the existence of token, option, and data. Hexadecimal format. Bit 0: The existence of token. Bit 1-6: The count of option. Bit 7: The existence of data.
<tkl>	Integer type The token value length
<token>	String type The token value. Hexadecimal format
<opt_cnt>	Integer type The count of option, it will show when set AT+COAPCFG="Showrspopt",1
<opt_name>	Integer type The option name, it will show when set AT+COAPCFG="Showrspopt",1
<opt_value>	String type The option value, it will show when set AT+COAPCFG="Showrspopt",1

<length>	Integer type
	The data length. The max length is 512 bytes
<data>	String type
	Receive data from server

4. Appendix

4.1 Power on inspection process

- (1) AT //judge whether the module is powered on successfully
- (2) AT+CFUN=1 //flight mode off
- (3) AT+CEREG? //judge the attachment status of PS domain. If the second parameter is 1 or 5, the attachment is normal

4.2 Interaction process with COAP server

- (1) AT+ECDNS="coap.me" //obtain the IP address of the heap server 134.102.218.18

- (2) AT+COAPCREATE=56830 //create a COAP client
- (3) AT+COAPOPTION=1,11,"test" //set URI path to "/ test"
- (4) AT+COAPSEND=0,2,134.102.218.18,5683,5,"hello" // send get data request to coap: // coap.me: 5683 / test
- (5) +COAPURC: "rsp",2,2.01,32070,7,POST OK //the data replied by coap.me is received
- (6) AT+COAPDEL //delete the COAP client

5. Version

Version	Data	Comments
Draft	2018-11-11	
A	2018-12-01	
V1.1	2019-07-01	Update AT commands
V1.2	2020-02-20	

6. About US

Shanghai EigenCOMM Technology Co.,Ltd is established in Feb 2017 in Zhangjiang Hi-Tech park, Shanghai China (www.eigencomm.com) . EigenCOMM focuses on cellular based IoT chipset and solution. With accumulation of experience over the years in algorithm, L1/L2/L3 protocol, SoC, RF , platform & application software as well as the communication architecture and system, team makes its way to IoT industry.

The ultra-low power NB-IoT chipset EC616(S) has already enter mass production stage. The 4G Cat.1 chipset EC618 is in engineering samples stage, 5G is also in the R&D stage.

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